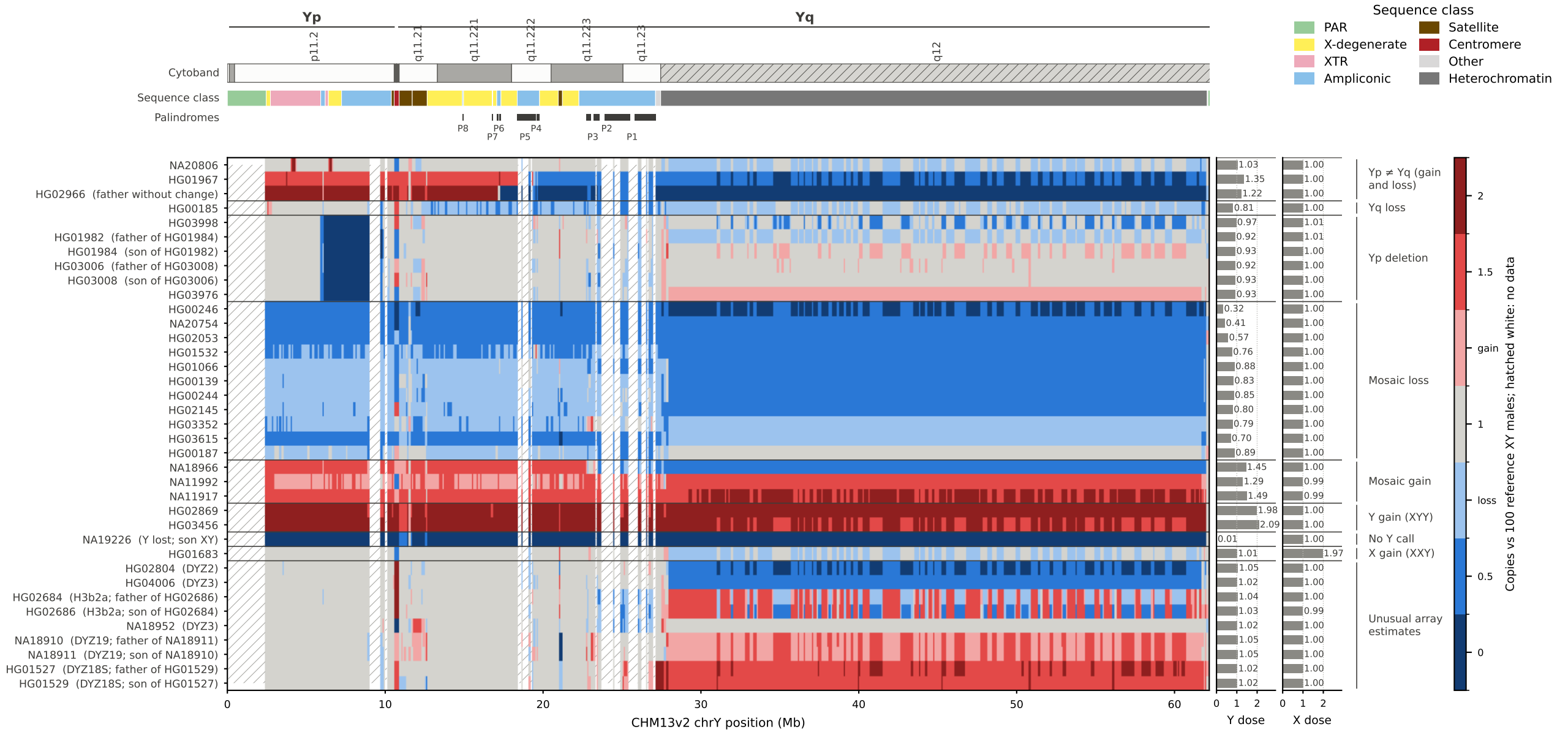

Heatmap: each 100 kb window of CHM13v2 chrY shows the coverage over the median of 100 reference XY males (Y state normal; 1 = as in the reference; 2 = twice). Where reads map uniquely, coverage is MQ20 reads per chromosome 2 read (GRCh38 alignments of the 1kGP CRAMs, moved to CHM13v2 with the UCSC hg38ToHs1 chain); in the arrays (centromere, DYZ19, Yq12), where reads do not map uniquely, it is the median count of the yruler marker k-mers that lie in the window. In Yq12 the value is the array amount relative to the median male, not a chromosome copy number. Colour steps: 0 (< 0.25), 0.5 (0.25-0.75), loss (0.75-0.9), 1 (0.9-1.1), gain (1.1-1.25), 1.5 (1.25-1.75), 2 (>= 1.75). Hatched white: no unique reads and fewer than 50 marker k-mers.
Y dose = 2 x mean count of the Y single-copy ruler k-mers / mean count of the chromosome 2 k-mers (Y against an autosome). X dose = chrX MQ20 reads outside the pseudoautosomal regions / chr2 MQ20 reads, over the median of males with Y dose 0.95-1.05.
Y state, first rule that applies: no Y call, Y dose <= 0.0222 (highest of 65 heldout females); Yp deletion, Yp dose < 0.5 and Yq dose within 0.1 of 1; Yq loss, Yp dose within 0.1 of 1 and Yq dose more than 0.1 below Yp; Yp ≠ Yq (state partial in the tables), Yp and Yq doses differ by more than 0.1, so part of the Y is gained and part lost; Y gain, Y dose >= 1.75; mosaic gain, 1.25 to 1.75; mosaic loss, < 0.9. XYY and XXY need both doses: XYY is Y gain with X dose 1, XXY is X dose 2 with Y dose 1 (X dose in 1,567 XY males with Y state normal: 0.981-1.022). Unusual array estimates: males noted in the cohort analysis. Rows are ordered by average-linkage clustering within each group; fathers and sons that are both in the figure are drawn next to each other.